Targeted SAT Math — Set 04

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Homework after Bluebook Practice Test 4. Work on paper. Calculator / Desmos is fine.

These are **new** questions in the same family as the Module 2 population-growth item — not that official question, and not the “find the months” copies from Set 01.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

Write the job first. Are they asking for the **percent**, the **months**, or the **exponent**?

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1. Notes for this set only

One cycle = exponent equals 1

A model $A(1+r)^{kt}$ finishes **one** growth (or decay) cycle when the exponent equals **1**.

- The **percent** n comes from the **base**. 1.08 means $n = 8$. It is not 1.08, and it is not a leftover number in the fraction.
- The **period** is the time until that exponent hits 1. If t is in years, convert that time into months when they ask for months: months $= 12 \times (\text{years per cycle})$.
- The coefficient k and the period **in years** are **reciprocals**. If one cycle takes $\frac{3}{2}$ years (18 months), then $k = \frac{2}{3}$.

They can hide any one piece

They give	They hide	What you do
The equation	The percent	Read n from the base. Then check that the period they named is when the exponent is 1.
The equation	The months	Set the exponent equal to 1 and convert.
The percent and the months	The exponent	Convert months → years. Then $k = \dfrac{1}{\text{years per cycle}}$.
t already in **months**	Either piece	Do **not** also divide by 12. One cycle is still “exponent $= 1$.”

Sometimes the exponent is written unsimplified, like $t/0.75$ instead of $\frac{4}{3}t$. Same clock. Set that expression equal to 1.

Decay uses a base less than 1. 0.82 means it **decreases** by 18% each cycle, not by 82%.

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2. Questions

1. The function P is defined by $P(t) = 420(1.12)^{\frac{3}{5}t}$, where t is the number of years after 2016. According to the model, the population increases by $n\%$ every 20 months. What is the value of n?

A) 3
B) 5
C) 12
D) 20

2. A bacteria culture increases by 8% every 18 months. The function C models the size of the culture t years after 2019. Which equation could define C?

A) $C(t) = 80(1.08)^{\frac{2}{3}t}$
B) $C(t) = 80(1.08)^{\frac{3}{2}t}$
C) $C(t) = 80(1.08)^{18t}$
D) $C(t) = 80(1.08)^{t/18}$

3. (SPR) The function R is defined by $R(m) = 60(1.06)^{m/10}$, where m is the number of **months** after January. According to the model, R increases by 6% every n months. What is the value of n?

4. (SPR) The function W is defined by $W(t) = 900(0.82)^{\frac{4}{5}t}$, where t is years after 2014. According to the model, the amount decreases by $n\%$ every 15 months. What is the value of n?

5. (SPR) A lake's algae count increases by 7% every 15 months. The count t years after 2021 is modeled by $A(t) = B(1.07)^{kt}$, where B and k are constants and $k > 0$. What is the value of k?

6. The function H is defined by $H(t) = 75(1.15)^{\frac{2}{5}t}$, where t is years after 2018. According to the model, H increases by $n\%$ every 30 months. What is n?

A) 2
B) 5
C) 15
D) 30

7. A wildlife count increases by 5% every 8 months. The count t years after 2017 is modeled by $F(t) = F_0(1.05)^{(\text{exponent})}$. Which expression could be the exponent?

A) $\dfrac{3}{2}t$
B) $\dfrac{2}{3}t$
C) $8t$
D) $\dfrac{t}{8}$

8. The function Q is defined by $Q(t) = 200(1.04)^{\frac{3}{2}t}$, where t is years after 2020. According to the model, Q increases by 4% every n months. What is the value of n?

A) 4
B) 8
C) 12
D) 18

9. The function N is defined by $N(m) = 140(1.09)^{m/8}$, where m is the number of **months** after March. According to the model, N increases by $n\%$ every 8 months. What is n?

A) 2
B) 8
C) 9
D) 91

10. The function S is defined by $S(t) = 110(1.10)^{t/0.75}$, where t is years after 2015. According to the model, the quantity increases by $n\%$ every 9 months. What is the value of n?

A) 0.75
B) 9

C) 10

D) 1.10